

Week 2 Portfolio Project

Part 1: Company Profile

1.1 Company Name

ABC Startup Solutions

1.2 Nature of Business

ABC Startup Solutions is a newly established software development company. It develops web and mobile applications, builds custom software systems for small and medium enterprises, and offers basic IT consulting services to its clients.

1.3 Company Vision

To be the leading provider of innovative software solutions that help Filipino businesses succeed in the digital age.

1.4 Company Mission

To deliver high quality software products and services that solve real business problems, create value for our clients, and provide a great working environment for our employees.

1.5 Office Location (Fictional)

2nd Floor, Veritas Corporate Building, Ayala Highway, Santa Rosa City, Laguna, Philippines.

1.6 Organizational Structure

Level	Department	Position	Count
Executive	Office of the CEO	Chief Executive Officer	1
Department	Information Technology	IT Manager, Junior System Administrator, Network Specialist, Software Developer, QA Support	5
Department	Human Resources	HR Manager, HR Associate, Recruitment Specialist, Admin Assistant	4
Department	Finance	Finance Manager, Accountant (2), Bookkeeper, Payroll Officer	5
Department	Sales	Sales Manager, Sales Executive (5)	6
Total			20

The company is headed by the Chief Executive Officer, who is also one of the founders. Each department is managed by a department head who reports directly to the CEO.

1.7 Employee Distribution

Department	Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
TOTAL	20

Part 2: Enterprise Hardware Inventory

The hardware below was selected based on the 20 employees, the work performed by each department, and the services that the company needs to run on its first day of operations.

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computer	12	Finance (5), HR (4), IT (3)	Handles daily office tasks such as reports, spreadsheets, records, and software development.
HW-002	Laptop	8	Sales (6), IT (2)	For employees who move around: sales visits clients and IT staff attend deployments and meetings.
HW-003	Server	1	IT (Server Room)	Domain controller and file server. Runs Ubuntu Server for internal services and web hosting.
HW-004	Router	2	IT (Server Room)	Main router for the network plus one spare unit for quick replacement in case of failure.
HW-005	Network Switch	3	IT (Server Room)	One 48-port main switch and two 24-port switches for the office floor and the server rack.
HW-006	Printer	2	HR, Sales	One monochrome laser printer for contracts and documents, one color printer for reports and marketing materials.
HW-007	UPS	6	Finance (5), IT (1)	Protects finance workstations and the server room against power interruptions. Finance data must not be lost.
HW-008	Wireless Access Point	2	Office Floor	Covers the entire office floor so laptops and mobile devices have a stable Wi-Fi connection.
HW-009	NAS Storage	1	IT (Server Room)	Central file storage with scheduled backups of company documents and project files.
HW-010	External Backup Drive	1	IT (Server Room)	Offline backup copy of the NAS, following the 3-2-1 backup rule.
HW-011	Monitor	8	Finance (5), IT (3)	Extra monitors for dual monitor setups. Finance staff check reports side by side and IT staff read code comfortably.

Justification of Quantities

The quantities are based on the number of employees per department. Finance and HR staff work mostly at their desks, so they receive desktop computers. Sales staff are often outside the office, so they receive laptops instead. The IT department gets a mix of both because some tasks, like server setup, are done on site while others, like deployment, are done at client locations.

The server and network devices are centralized in the server room. A single server is enough for a 20-person company since it will only run a domain controller, a file server, and internal web services. The two routers follow the basic rule that critical network devices should have a spare. The UPS units protect the most critical workstations: the finance department, whose records must always be available, and the server room, which must never lose power unexpectedly.

Dual monitors are limited to the departments that really need them. Finance employees compare spreadsheets side by side and IT employees write and test code, so giving them extra monitors increases their productivity without spending money on devices that sales and HR do not actually need.

Part 3: Enterprise Software Inventory

The software below covers the operating systems, productivity tools, development tools, and utilities needed by the company.

Software	Version	License	Purpose
Windows 11 Pro	24H2	OEM (with hardware)	Operating system for all desktops and laptops. The Pro edition is needed for domain join, BitLocker encryption, and Remote Desktop.
Ubuntu Server	24.04 LTS	Open Source (GPL)	Runs the internal web server, file services, and testing environments. Free, stable, and widely used in production.
Microsoft 365 Apps	Latest	Subscription (per user)	Word, Excel, PowerPoint, and Outlook for office documents, financial reports, and company email.
Visual Studio Code	1.105	Free (MIT)	Main code editor of the IT team for web and mobile application development.
Git	2.48	Open Source (GPL)	Version control for all source code so the team can track changes and work together safely.
GitHub Desktop	3.4	Free	User friendly Git interface for team members who are not yet comfortable with the command line.
VirtualBox	7.1	Open Source (GPL)	Creates virtual machines for testing servers and applications without buying extra hardware.
Google Chrome	140	Freeware	Main browser for daily web use, development, and testing.
Microsoft Defender	Built-in	Included with Windows	Baseline antivirus and malware protection on all company computers.
AnyDesk	9.1	Freemium	Remote support tool so the IT team can fix employee computers without leaving their seats.
7-Zip	24.09	Open Source (LGPL)	Compresses and extracts files, and handles large log files and backups.

Why Each Software Is Needed

Windows 11 Pro is the standard operating system for the workstations because the business versions require domain joining, which allows the IT department to manage all computers centrally. Ubuntu Server is used for the company's internal servers because it is free, stable, and supported for a long period, which keeps licensing costs low.

Microsoft 365 covers the office needs of every department: documents, spreadsheets, presentations, and email. The development stack of the company is built on free tools: Visual Studio Code for editing code, Git and GitHub Desktop for version control, and VirtualBox for testing. Google Chrome is the standard browser because it works well for both development and everyday use.

Microsoft Defender provides the baseline antivirus protection for free since it is built into Windows. AnyDesk gives the IT team remote access for troubleshooting, and 7-Zip is used for compressing backups and transferring large files. All of these are either free or already included in the hardware purchase, which is important for a startup budget.

Part 4: Enterprise Network Inventory

The table below lists the networking equipment required to build the company network from scratch.

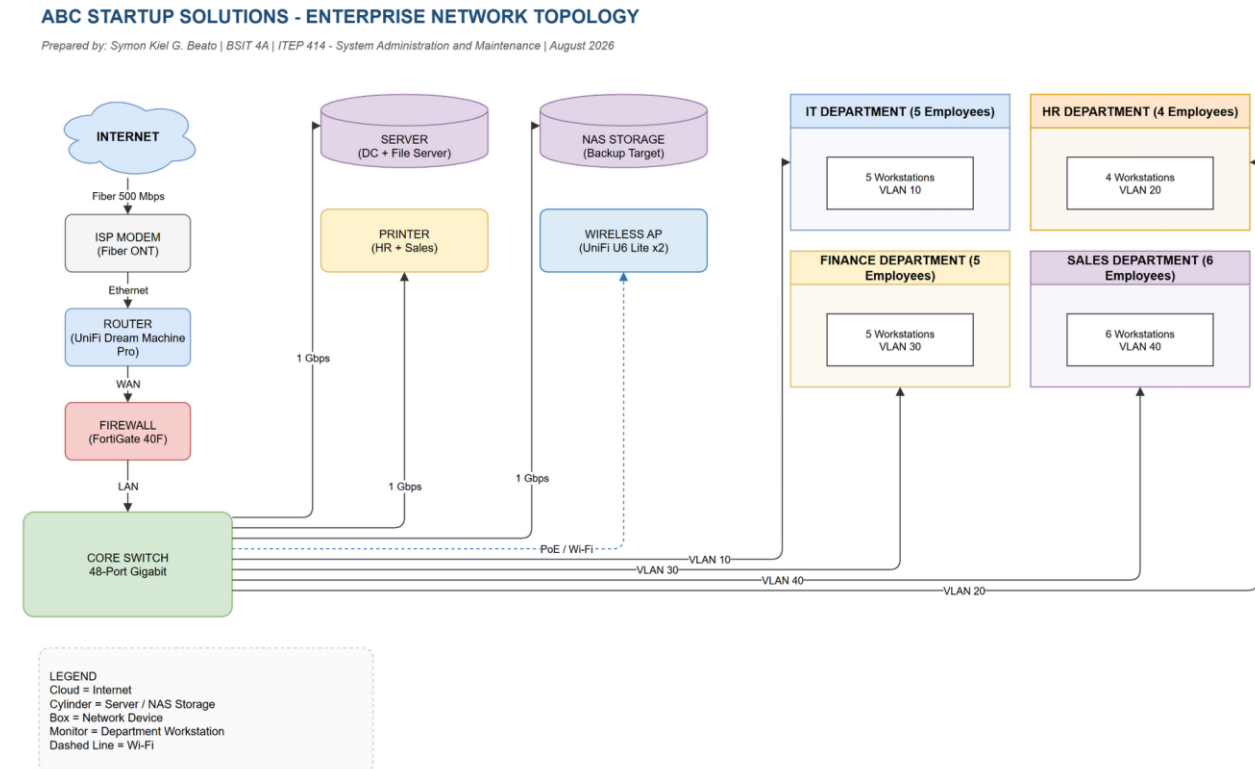
Item	Specification	Quantity	Location	Purpose
ISP Modem / ONT	Fiber ONT provided by the ISP	1	Server Room	Converts the fiber optic signal from the internet provider into an Ethernet connection.
Router	Ubiquiti UniFi Dream Machine Pro	1	Server Room	Main gateway of the network. Handles routing, DHCP, and VLANs.
Firewall	FortiGate 40F	1	Server Room	Blocks unwanted traffic and secures the boundary between the internet and the company network.
Switch (Core)	48-port managed Gigabit switch	1	Server Room	Central switch that connects the server, access points, and department switches.
Switch (Floor)	24-port managed Gigabit switch	2	Office Floor	Provides wired connections for the workstations of each department.
Access Point	UniFi U6 Lite	2	Ceiling, Office Floor	Provides Wi-Fi coverage for laptops, phones, and guest devices.
Patch Panel	24-port CAT6	2	Server Rack, Office	Clean and organized termination point for the network cables.
CAT6 Cable	305-meter boxes	3	Office Floor	Running network cables from the rack to every workstation and wall outlet.
RJ45 Connectors	CAT6 rated	100	Office Floor	Terminating the ends of network cables to the patch panels and wall outlets.

The network was designed for easy expansion. The 48-port core switch can already support more than 60 devices, which gives the company room to grow before new equipment is needed. A separate firewall was chosen instead of relying only on the router's built-in filtering, because every software company should treat its perimeter security as a dedicated layer. All cabling is CAT6, which supports Gigabit speeds and is the standard for office networks today.

Part 5: Enterprise Network Diagram

The network topology was designed using Draw.io. It shows the flow of data from the internet down to the workstations of the four departments.

Figure 1: ABC Startup Solutions Enterprise Network Topology



Note: The diagram was created in Draw.io and exported as PNG and PDF from the file diagrams/NetworkDiagram.drawio.

How the Network Flows

- The ISP Modem receives the fiber internet connection from the internet service provider.
- The Router receives the signal and handles IP addressing and routing between networks.
- The Firewall filters all traffic before it enters the company network.
- The Core Switch distributes the connection to the server, NAS, printer, wireless access point, and the four departments.
- The Wireless Access Point provides Wi-Fi for laptops and mobile devices.
- Each department is connected to the switch through its own VLAN: VLAN 10 for IT, VLAN 20 for HR, VLAN 30 for Finance, and VLAN 40 for Sales.

Part 6: System Administration Roles

This section discusses four positions that exist inside a typical IT organization and explains how they work together.

6.1 Helpdesk Technician

The Helpdesk Technician is the first person that employees talk to when something goes wrong with their computer, email, or software.

Responsibilities

- Receives support requests through tickets, phone calls, and walk-ins.
- Troubleshoots common hardware and software issues such as login problems, printer errors, and software installations.
- Escalates issues that require higher level support to the network or system administrators.
- Keeps ticket records updated and writes simple guides for the knowledge base.

Skills

- Good communication and customer service.
- Patience and the ability to explain technical issues in simple terms.
- Basic knowledge of Windows, office software, and common peripherals.
- Logical troubleshooting and note taking.

Common Tools

- Ticketing systems such as Zendesk and ServiceNow.
- Remote support tools such as AnyDesk and TeamViewer.
- Active Directory for password resets and account locks.

Certifications

- CompTIA A+
- ITIL Foundation
- Microsoft Fundamentals certifications

Source: <https://www.comptia.org/certifications/a>

6.2 Network Administrator

The Network Administrator designs, installs, and maintains the network that connects all company devices.

Responsibilities

- Configures and maintains routers, switches, firewalls, and wireless access points.
- Monitors network performance, uptime, and bandwidth usage.
- Manages IP addresses, VLANs, and VPN connections for remote access.

- Documents the network and fixes connectivity issues when they happen.

Skills

- Strong understanding of TCP/IP, routing, and switching.
- Knowledge of network security fundamentals and firewall rules.
- Experience with cabling, patching, and network equipment.
- Patience for long troubleshooting sessions.

Common Tools

- Wireshark for packet analysis.
- PRTG or SolarWinds for network monitoring.
- PuTTY and console access for device configuration.
- Nmap for network scanning and checking open ports.

Certifications

- CompTIA Network+
- Cisco CCNA
- Juniper JNCIA

Source: <https://www.cisco.com/site/us/en/learn/training-certifications/certifications/enterprise/ccna/index.html>

6.3 Linux System Administrator

The Linux System Administrator manages the servers that keep company services running, which in this company are based on Ubuntu Server.

Responsibilities

- Installs and maintains Linux servers for web hosting, file services, and internal applications.
- Manages user accounts, permissions, and system services.
- Writes shell scripts to automate repetitive tasks.
- Applies security patches and keeps servers updated.

Skills

- Comfort with the command line and shell scripting in Bash.
- Knowledge of service management with systemd.
- Understanding of filesystems, permissions, and basic networking.
- A careful approach to changes, since one mistake can take a server offline.

Common Tools

- SSH for remote server access.
- Ansible for configuration management.
- Docker for running applications in containers.
- System tools such as systemctl, cron, rsync, htop, and vim.

Certifications

- LPIC-1 and LPIC-2
- Red Hat RHCSA
- CompTIA Linux+

Source: <https://www.redhat.com/en/services/training/rhcsa-red-hat-certified-system-administrator>

6.4 Cloud Administrator

The Cloud Administrator manages the company's resources hosted on cloud platforms such as AWS, Azure, or Google Cloud.

Responsibilities

- Creates and manages cloud servers, storage, and databases.
- Monitors cloud spending and keeps costs within budget.
- Configures backups, monitoring, and automatic scaling.
- Implements cloud security such as IAM roles, access keys, and policies.

Skills

- Working knowledge of at least one major cloud platform.
- Understanding of identity and access management.
- Familiarity with infrastructure as code.
- Cost awareness and the habit of checking usage reports.

Common Tools

- AWS Console and Azure Portal.
- Terraform and CloudFormation for infrastructure as code.
- AWS CLI for scripting and automation.
- CloudWatch and Azure Monitor for alerts.

Certifications

- AWS Certified SysOps Administrator
- Microsoft Azure Administrator (AZ-104)
- Google Cloud Associate Cloud Engineer

Source: <https://aws.amazon.com/certification/certified-sysops-admin-associate/>

How These Four Professionals Work Together

Inside one organization, these four roles work together like the parts of one system. When an employee reports a problem, the Helpdesk Technician receives the ticket first and does the initial troubleshooting. If the issue is on the network side, it is passed to the Network Administrator. If a server is slow or down, the Linux System Administrator investigates, and if the affected service is hosted in the cloud, the Cloud Administrator handles it. They share the same monitoring dashboards and documentation, so when an incident happens, everyone knows which part of the infrastructure is their

responsibility. For example, if a sales employee cannot open the company web application, the Helpdesk Technician checks the device, the Network Administrator checks the connection, the Cloud Administrator checks the cloud server, and the Linux Administrator checks the application logs. In short, the Helpdesk Technician is the front door, the Network Administrator keeps the roads open, the Linux System Administrator runs the building, and the Cloud Administrator manages the branch office.

Part 7: Infrastructure Recommendations

Internet Provider

The company should subscribe to a business fiber plan from Converge or PLDT with 500 Mbps download and upload speed. A software development company needs a strong upload speed because it pushes large code files, holds video meetings, and uploads backups to the cloud. A business plan is preferred over a residential plan because it comes with a static IP address, which is needed for VPN access and remote administration, and it includes faster support in case of outages.

Server Specifications

A single on-premise server is enough at this stage: a Dell PowerEdge T350 with an Intel Xeon processor, 32 GB ECC RAM, two 1 TB SSDs in RAID 1 for the operating system and services, and two 4 TB HDDs in RAID 1 for file storage. ECC RAM is important for a server because it detects and corrects memory errors that could corrupt data. RAID 1 protects the company against a single disk failure without any downtime.

Backup Strategy

The company should follow the 3-2-1 backup rule: three copies of the data, stored on two different types of media, with one copy kept offsite. Daily backups go to the NAS, weekly backups go to the external backup drive, and monthly backups are uploaded to a cloud service such as Backblaze. The external drive should be kept in a fireproof cabinet or brought home by the system administrator. A restore test should be performed every quarter to make sure the backups actually work.

Security Recommendations

The FortiGate firewall should be configured with a default-deny policy so that only approved traffic passes through. Unused ports and services must be disabled, and all software should be updated monthly. Access should follow the principle of least privilege: employees only get access to the files and systems they need for their work. The server room should have a lock and an access log. Employee security training should be conducted once a quarter, since most security incidents start with a human mistake such as clicking a phishing link.

Antivirus

Microsoft Defender is a good free baseline because it is already built into Windows and provides real-time protection. For stronger protection against ransomware and advanced threats, the company should add a business endpoint solution such as Bitdefender GravityZone or SentinelOne on all workstations and the server. Definitions should update automatically, and a full scan should run weekly.

Password Policy

Passwords must be at least 12 characters long and use a mix of letters, numbers, and symbols. Multi-factor authentication (MFA) should be enabled on all company accounts, especially email and

cloud accounts. Sharing passwords is not allowed, and every employee should use a password manager such as Bitwarden to keep track of their accounts. Instead of forcing password changes every 60 or 90 days, which pushes users to pick weak passwords, passwords should only be changed when a compromise is suspected.

Expansion Plan

The network was designed with room to grow. The 48-port core switch can support up to 60 devices, so the company can hire up to 40 employees before a new switch is needed. When the team grows beyond that, a second floor switch and a second server for redundancy should be added. Public-facing web hosting should eventually move to the cloud on AWS or Azure, keeping the on-premise server only for internal files. When the infrastructure becomes too big for one person to manage, the company should hire a dedicated Network Administrator.

Part 8: Personal Reflection

Before this project, I thought being a System Administrator was mostly about fixing computers and resetting passwords. After working on this portfolio, I realized that the role starts long before the first computer is even bought. The most important lesson I learned is that infrastructure planning must begin with the business. A company of 20 people has different needs from a company of 200, and every decision, from the size of the server to the number of switches, should trace back to the actual number of employees and the work they do.

The task I found most challenging was preparing the hardware inventory. At first I just wanted to list down the items, but the requirement to justify every quantity forced me to think like a real administrator. Why does the Finance department need UPS units? Because their work cannot afford to be interrupted by a power outage. Why 12 desktops and 8 laptops? Because most employees stay at their desks while the sales team moves around. These small decisions showed me that a good inventory is not a shopping list. It is a document that explains the logic of the whole infrastructure. The network diagram was also difficult, especially tracing the flow of data from the internet down to each department, but it became clearer once I imagined the network as a road system.

This project also taught me why planning must come before deployment. Buying equipment without a plan wastes money and creates problems that are expensive to fix later. A missing switch port or a server that is too small means redoing the setup and spending again. Planning on paper is cheap, but fixing mistakes after everything is installed is not. I also learned the value of documentation. The README file, the inventories, and the diagram are not just requirements. They are tools that the next administrator, or even my future self, will use to understand the network without asking around.

In the end, this project gave me a clearer picture of what a System Administrator really does and how much thinking goes into a simple office network. It helped me practice attention to detail, technical writing, and problem solving. I feel more confident about facing real infrastructure tasks during my internship, and I now understand why employers value these planning skills. I am excited to apply what I learned, and I plan to keep building on this portfolio as I learn more.

References

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